

Itinerant Magnetism

SS25 Lecture Notes

July 15, 2025

2025-07-15

Itinerant Magnetism

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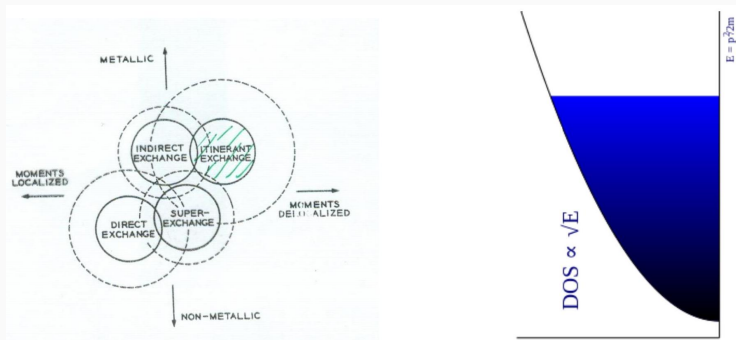
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V. Itinerant magnetism

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- Crossover from localized to delocalized (itinerant) magnetism.
 - Metal-to-insulating crossover still an open problem.
- Open problems: cuprates, FeAs/FeSe, heavy-fermion, 4d/5d-oxides

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Open problems: cuprates, FeAs/FeSe, heavy-fermion, 4d/5d-oxides

Characteristic parameters

iron, cobalt, nickel very often metallic and rather itinerant iron 2.12 B, 1044 K nickel 0.54 B ZrZn 0.12 B MnSi 0.4 B these numbers already hint that something more complicated than a simple local-moment picture is at work.

substance	μ_{sat} (μ _B)	μ_{eff} (μ _B)	T_{C} (K)	$\mu_{\text{sat}}/\mu_{\text{eff}}$	struct.	type
α-Fe	2.12	1.517	1044	1.01	bcc	FM
Co	1.66	1.618	1390	1.28	hcp	FM
Ni	0.54	2.1	624	1.375	fcc	FM
ZrZn	0.12	1.015	2127	1		WIF
MnSi	0.4	1.03	29	5.3	cubic	WIF

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V.1 Localmoment theories

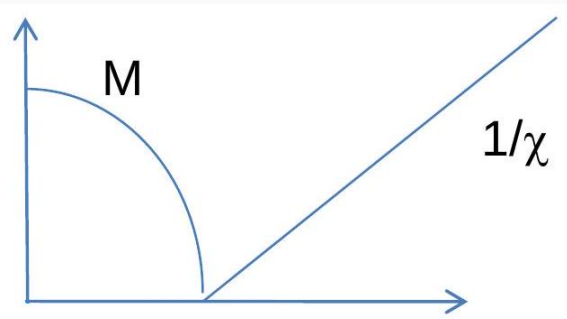
Langevin: fixed moment m

$$\langle m_z \rangle = mL(x) = m \left[\coth x - \frac{1}{x} \right], \quad x = \frac{mH}{k_B T}$$

Susceptibility: $\chi = \frac{C}{T}$ (Curie).

Weiss molecular field:

$$\langle m_z \rangle = mL\left(\frac{H + f\langle m_z \rangle}{k_B T}\right) \Rightarrow T_C = \frac{m^2 f}{3k_B}$$

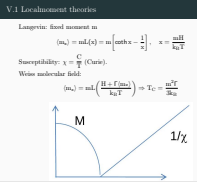


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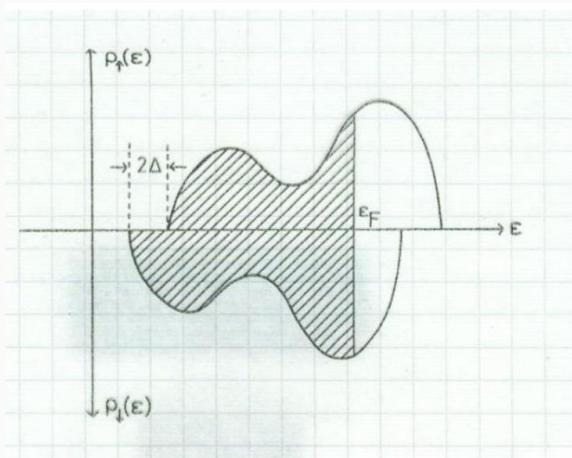
V.1 Localmoment theories

We already looked at the Langevin function magnetic moment of fixed size
Weiss molecular field you get a CurieWeiss law above T_C replace Langevin by Brillouin for quantum spins justified for local atomic moments (magnetic insulators, rare-earth compounds).



V.2 Itinerant (Stoner) picture

- Ferromagnetism in an itinerant electron system (Bloch, Wigner).
- Intra-atomic exchange decisive for d-electrons.
- Exchange energy: $I = J/N_0$.



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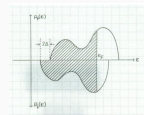
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└ V.2 Itinerant (Stoner) picture

We start from the band structure intra-atomic exchange (Hunds rule) Stoner
1938: split spin-up / spin-down bands criterion $I\rho(\epsilon_F) > 1$

V.2 Itinerant (Stoner) picture

- Ferromagnetism in an itinerant electron system (Bloch, Wigner).
- Intra-atomic exchange decisive for d-electrons.
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Stoner theory

Band splitting lowers exchange energy but raises kinetic energy:

$$E_{\text{ex}} = \frac{1}{4}IN^2 - IM^2, \quad E_{\text{kin}} \approx \frac{M^2}{\rho(E_F)} + \dots$$

Ferromagnetism when

$$\alpha_0 \equiv I\rho(E_F) > 1$$

Problems:

- Predicts T_C too high.
- Susceptibility not CurieWeiss above T_C .

Itinerant Magnetism

└ Stoner theory

Total energy = kinetic + exchange $\alpha_0 \equiv I\rho(E_F)$ must exceed 1 but calculated Curie temperatures are too high, and above T_C we do not obtain a CurieWeiss law.

Band splitting lowers exchange energy but raises kinetic energy.

$$E_{\text{ex}} = \frac{1}{4}IN^2 - IM^2, \quad E_{\text{kin}} \approx \frac{M^2}{\rho(E_F)} + \dots$$

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V.3 Localized vs. itinerant

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V.3 Localized vs. itinerant

- Localized (real space)
- Itinerant (reciprocal space)
- Anderson: Mott insulator $\Rightarrow$ localized d-electrons via super-exchange
- Metals: Fermi surface exists.
- How to reconcile?
- 1. Improve Stoner (include correlations).
- 2. Local moments in metals (van Vleck, Friedel).

Localized moments live in real space; itinerant electrons are delocalised, but can be localised in k-space In magnetic insulators d-electrons are localised by the Mott mechanism, while in metallic compounds a Fermi surface exists.

Weak itinerant ferromagnets

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Weak itinerant ferromagnets

$$\text{ZrZn}_2 \quad T_C = 25 \text{ K} \quad m = 0.12 \mu_B$$

$$\text{Sc}_2\text{In} \quad T_C = 6 \text{ K} \quad m = 0.04 \mu_B$$

- Ordered moment tiny, yet perfect CurieWeiss law!
- Effective moment $\gg$ ordered moment.

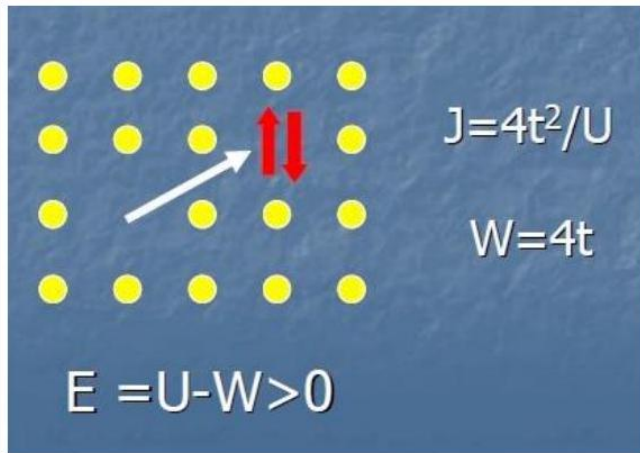
1960 Matthias et al.: ZrZn 25 K, 0.12 B; ScIn 6 K, 0.04 B These findings strongly disagree with any local-moment or Heisenberg picture, yet they exhibit a perfect CurieWeiss law with an effective moment much larger than the ordered moment.

$$\begin{array}{l} \text{ZrZn}_2 \quad T_C = 25 \text{ K} \quad m = 0.12 \mu_B \\ \text{Sc}_2\text{In} \quad T_C = 6 \text{ K} \quad m = 0.04 \mu_B \end{array}$$

- Ordered moment tiny, yet perfect CurieWeiss law!
- Effective moment $\gg$ ordered moment.

V.4 Mean-field (Hubbard) model

$$H = \sum_{i,j} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$



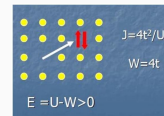
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$$H = \sum_{i,j} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$



The Hubbard model: kinetic term + on-site Coulomb U Atom: $U = 10$ eV,
screened $I = 1$ eV

Ferromagnetism within Hubbard

Split bands: $\epsilon_{k\sigma} = \epsilon_k + \sigma\Delta$, $\Delta = IM + \mu_B H$.

Free-energy expansion:

$$F(M, T) = F_0 + \frac{M^2}{2\chi_0(T)} + \frac{g(T)}{4}M^4 + \dots = \frac{1}{2}IM^2 - IM$$

Results:

- $T_C \propto \sqrt{\alpha_0 - 1}$,
- $M(T) = M(0)\sqrt{1 - (T/T_C)^2}$,
- No CurieWeiss $\chi(T)$.

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- No CurieWeiss $\chi(T)$.

Stoner criterion again: $I\rho(E_F) > 1$ Magnetisation $M(T) = M(0)\sqrt{1 - (T/T_C)^2}$ (disagrees with Bloch $T^{3/2}$) Above T_C no CurieWeiss law.

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└ V.4.2 Antiferromagnetism

Separate into sub-lattices A, B:
 $M_{\text{staggered}} = M_A - M_B, \quad F = \frac{1}{2} \left(\frac{1}{N_A} - 1 \right) M_A^2 + \dots$
Condition: $2\chi_0^2(T_N) = 1$.
Susceptibility depends on wave-vector $\vec{Q}$:
 $\chi_0(\vec{Q}) = \frac{1}{4} \sum_{\vec{r}} \frac{\vec{S}_i \cdot \vec{S}_{i+\vec{Q}}}{r_{i+\vec{Q}} - r_i}$

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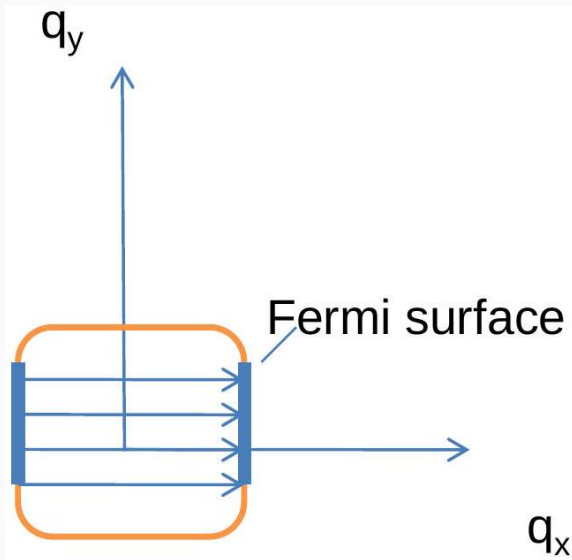
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$$\chi_0(\vec{Q}) = \frac{1}{4} \sum_{\vec{r}} \frac{\vec{S}_i \cdot \vec{S}_{i+\vec{Q}}}{r_{i+\vec{Q}} - r_i}$$

Slater: AFM also possible in itinerant electron model Staggered magnetisation, staggered susceptibility $\chi(\vec{Q})$

Spin-density waves (SDW)

Fermi-surface nesting $\Rightarrow$ large $\chi_0(\vec{Q})$

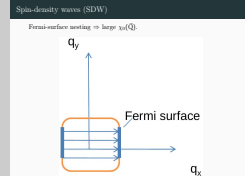


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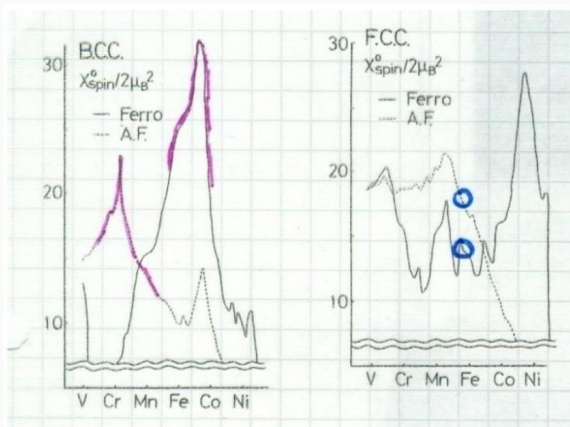
└ Spin-density waves (SDW)

Nesting: flat parallel parts of the Fermi surface give a large Lindhard response
SDW order when $2I\chi_0(\vec{Q}) > 1$



Which magnetic order?

- Depends on Fermi-surface shape.
- FM and AFM can coexist (e.g. Fe: FM bcc, AFM fcc).

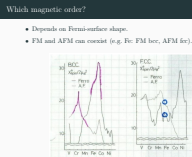


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└ Which magnetic order?

Example Fe: bcc FM, fcc AFM Stainless steel (mostly fcc) is non-magnetic consequence of the phase diagram.



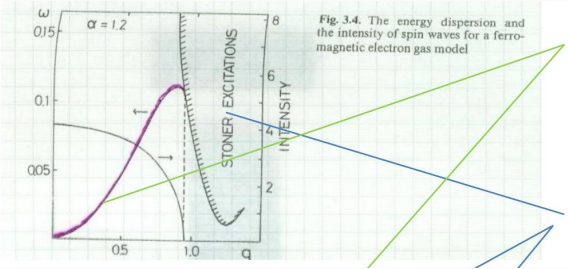


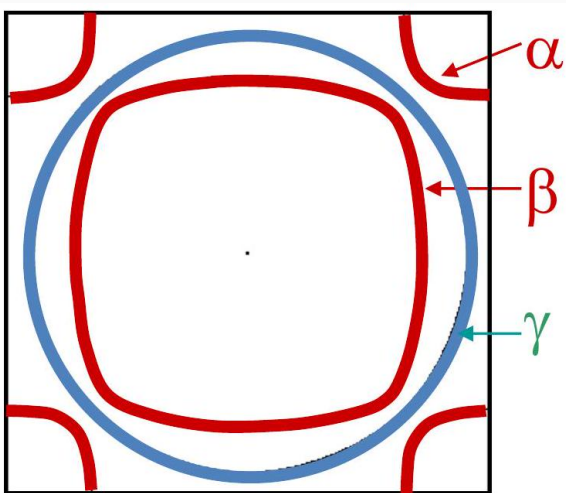
Fig. 3.4. The energy dispersion and the intensity of spin waves for a ferromagnetic electron gas model

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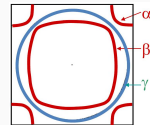
V.5 Dynamical mean-field (RPA)

RPA: electron-hole pair excitations + exchange enhancement Stoner continuum and spin-wave (magnon) bound states Still misses thermodynamics; SCR / paramagnon theories needed.

$$\chi(\vec{Q}, \omega) = \frac{\chi_0(\vec{Q}, \omega)}{1 - 2\chi_0(\vec{Q}, \omega)}$$



└ Fermi-surface nesting in Sr_2RuO_4

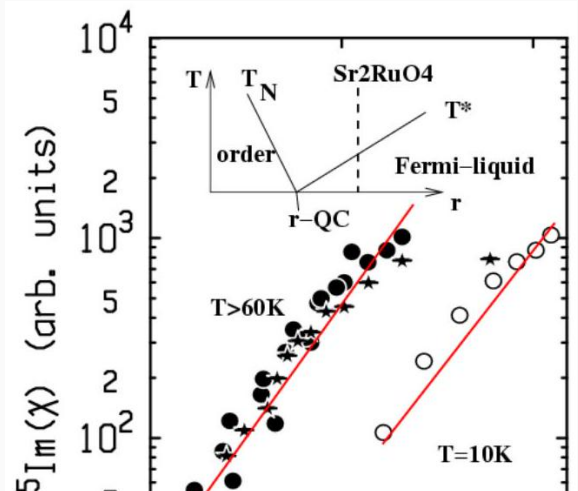


Incommensurate fluctuations due to nesting in quasi-1D bands Measured by inelastic neutron scattering.

EnergyTemperature dependence

$$\chi''(q_0,0) = \chi''(q_0,0) \frac{\Gamma\omega}{\Gamma^2 + \omega^2}$$

Parameters χ' , Γ track temperature $\Rightarrow$ close to instability.

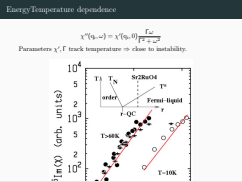


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└ EnergyTemperature dependence

$\chi'(q_0,0)$, Γ and FWHM vary with T all indicate proximity to a quantum-critical instability.

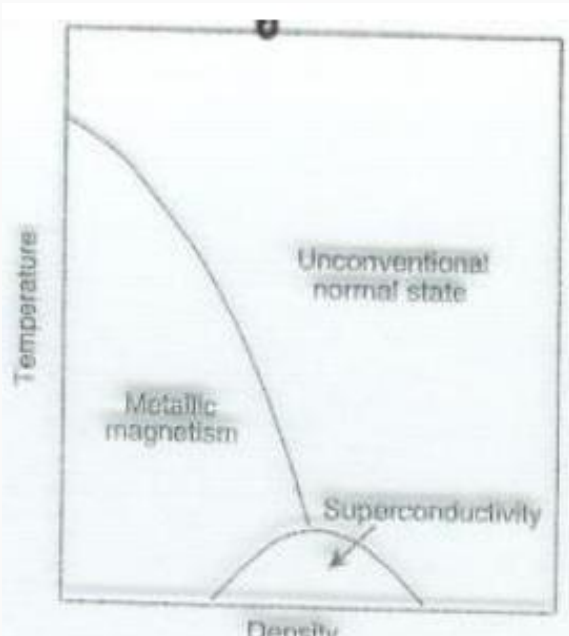


└ Influence of spin fluctuations

- Nearly FM metal: $R - R_0 \propto T^{5/3}$ (FM phase T^2).
- Nearly AFM metal: $R - R_0 \propto T^{3/2}$.

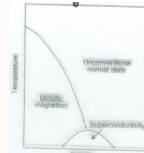
- Nearly FM metal: $R - R_0 \propto T^{5/3}$ (FM phase T^2).
- Nearly AFM metal: $R - R_0 \propto T^{3/2}$.

Spin-fluctuation contributions to resistivity: FM side $T^{5/3}$, AFM side $T^{3/2}$
SCR theory quantitatively describes these power laws.



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└ V.6 Magnetism & Superconductivity



SC often emerges when magnetic order is suppressed by pressure or doping
quantum-critical scenario Examples: Ce- and U-based systems, cuprates, Fe-pnictides.

General

- C. Kittel – Introduction to Solid State Physics
- IbachLüth – Festkörperphysik
- AshcroftMermin – Solid State Physics

Magnetism

- P. Fazekas – Lecture Notes on Electron Correlation and Magnetism
- T. Moriya – Spin Fluctuations in Electron Magnetism
- MarshallLovesey – Theory of Thermal Neutron Scattering

Heres a list of standard and specialised literature.

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